

CSA1507 - CLOUD COMPUTING AND BIG DATA ANALYTICS

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1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

CLOUD COMPUTING: cloud computing means delivery of computing service such as server, storage, database, Analytics, networking, and software over the internet as on pay as you go basis.

Characteristics of cloud computing:-

→ **Rapid elasticity:** you can increase or decrease the service

- Elasticity - we can increase or decrease the service as required without any hassle.

→ **On Demand self service:**

- We can access the services (Required)
- we can self access the services we need.

→ **Resource pooling:**

- here we don't need physical services. we can access any service without having any physical service within.
- Due to this we don't need own the physical service we can just access the services through internet.
- here with one physical service multiple users can use.

→ **Broad Network Access**

- we can access any resource over the internet without any hassle.

3. Explain the role of server virtualization in cloud computing.

Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Server virtualization:

- server virtualization divides one physical server into multiple virtual server using hypervisor.
- each virtual machine runs its own operating system.

virtualization in cloud computing.

- virtualization is a technology that enables cloud computing.
- without virtualization cloud providers cannot share services through internet.

virtualization	cloud computing.
→ it divides one physical server into multiple virtual server → using hypervisor. → efficient use of hardware resource → usually implemented within on or more organization	→ It provides/delivery of computing service such as server, storage, database, network, software and analytics as pay as you go service → uses virtualization along with network, automation and management system

→ services are accessed through the internet provided by the service provider.

→ without virtualization cloud providers need to have separate physical server for each.

VIRTUALIZATION:-

- efficient use of hardware. - multiple VM run on single server
- cost Reduction. - fewer physical server required.
- scalability - we can reduce or increase the server based on requirement.
- Isolation - each has its own operating system and acts independently.
- Easy management - we don't need to maintain physical server we can access bc of this easily we can

Differentiate IaaS, PaaS, SaaS, XaaS with one suitable real-world example.

IaaS - ^{Infrastructure} ~~Internet~~ as a service.

PaaS - ^{Platform} ~~Platform~~ as a service

SaaS - software as a service

XaaS - Anything as a service

	Explanation.	Example
IaaS	provider virtual server, storage and network over the internet	AWS
PaaS	provides a platform for developing, testing and deploy application without managing hardware	Google cloud Google App Engine
SaaS	Provides ready to use software that user access through web browser.	Email, Google cloud Microsoft 365
XaaS	Refer to delivery of any IT services over the internet including storage	Microsoft 365

4. Discuss the role of data center in delivering cloud services.

Data center:

A Data center is a centralized facility that houses physical servers, storage devices, networking equipment, cooling systems, Power supply and security.

→ Data centers are called as back bone of cloud computing
Because Data centers are ~~are~~ manufacture of cloud services

1. Storage of Data:-

It provides large scale storage system to store user details Data or organization data.

→ It is important to have Data center because without Data center we cannot store user Data.

→ Having user Data is important - here the Data center holds user Data and gives whenever the user requires.

2. Hosting Application:

→ cloud applications are hosted on servers located inside the Data centers.

→ Data center manages the Data.

→ It acts like a house for all physical servers, storage devices, networking equipments.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud computing.

→ cloud computing - delivers computing services such as servers, storage, database, Analytics, networking, and software over the internet on a pay as you go basis.

Evolution of cloud computing:-

1. hardware: It is the first step. here people used to access servers through the physical server. you have to own the necessary servers to access them.

→ It is not cost efficient. you need to maintain the services physically based requirement.

2. ~~cloud server~~ client server:-

here people used to access the services based on client server. It is like a one to one service.

3. Internet service.

Internet enables users to access applications and services over the internet.

4. Virtualization:-

- It divides one physical server into multiple virtual servers using hypervisor.
- each has its own OS.
- efficient use of hardware resources.

5. Cloud computing.

- delivery of computing services such as server, storage, database, network, software and analytics as pay as you go service.
- services are accessed through the internet provided by the service provider.